

Lab Assignment 5

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Question1: Write a C program to reverse an integer/float array.

```
#include<stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    int i;
```

```
    printf("\nWhat lenght array do you want? ");
```

```
    scanf("%d", &n);
```

```
    int arr[n];
```

```
    for(i = 0; i < n; i++){
```

```
        printf("\n %d th element: ", i);
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    printf("\n The original array is: ");
```

```
    for(i = 0; i < n; i++){
```

```
        printf("%d \n", arr[i]);
```

```
    }
```

```
    for(i = 0; i < n/2; i++){
```

```
        int temp = arr[i];
```

```
        arr[i] = arr[n - 1 - i];
```

```
        arr[n - 1 - i] = temp;
```

```
    }
```

```
    printf("\n The original array is:\n ");
```

```
    for(i = 0; i < n; i++){
```

```
        printf("%d \n", arr[i]);
```

```
    }
```

```
}
```

```
atharva@quasar-2 Lab Assignment 5 % gcc reverse.c -o reverse
atharva@quasar-2 Lab Assignment 5 % ./reverse
```

```
What lenght array do you want? 5
```

```
0 th element: 26
```

```
1 th element: 96
```

```
2 th element: 49
```

```
3 th element: -50
```

```
4 th element: -349
```

```
The original array is: 26
```

```
96
```

```
49
```

```
-50
```

```
-349
```

```
The original array is:
```

```
-349
```

```
-50
```

```
49
```

```
96
```

```
26
```

```
atharva@quasar-2 Lab Assignment 5 % 
```

Question2: To insert an element at a given location of an array.

```
#include<stdio.h>
int main(){
    int n;
    int i;
    int index;
    int val;

    printf("\nEnter size of array: ");
    scanf("%d", &n);

    int arr[n+1];

    printf("\nEnter the elements: ");
    for ( i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }

    printf("\nOriginal Array: ");
    for( i = 0; i < n; i++){
        printf("%d ", arr[i]);
    }

    printf("\n");

    printf("At which index do you want the new value? ->");
    scanf("%d", &index);

    printf("\nEnter new element: ");
    scanf("%d", &val);

    for ( i = n; i > index; i--){
        arr[i] = arr[i-1];
    }
    arr[index] = val;
    n++;

    printf("\nUpdate Array\n");
    for(i = 0; i < n; i++){
        printf("%d ", arr[i]);
    }
}
```

```
atharva@quasar-2 Lab Assignment 5 % gcc insert.c -o insert
atharva@quasar-2 Lab Assignment 5 % ./insert
```

```
Enter size of array: 5
```

```
Enter the elements: 12 34 06 44 19
```

```
Original Array: 12 34 6 44 19
```

```
At which index do you want the new value? ->4
```

```
Enter new element: 23
```

```
Update Array
```

```
12 34 6 44 23 19 %
```

```
atharva@quasar-2 Lab Assignment 5 %
```

Question3: Delete a given element in an array.

//Doubt - desired output is not coming!!

// reused the code written for insertion as deletion is opposite of insertion

```
#include<stdio.h>
int main(){
    int n;
    int i;
    int index;
    // int val;

    printf("\nEnter size of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("\nEnter the elements: ");
    for ( i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }

    printf("\nOriginal Array: ");
    for( i = 0; i < n; i++){
        printf("%d ", arr[i]);
    }
    printf("\n");

    printf("From which index do you want to delete? ->");
    scanf("%d", &index);

    // printf("\nEnter new element: ");
    // scanf("%d", &val);

    for ( i = index; i < n - 1; i++){
        arr[i] = arr[i+1];
    }

    // arr[index] = val;
    n--;

    printf("\nUpdate Array\n");
```

```
    for(i = 0; i < n; i++){  
        printf("%d ", arr[i]);  
    }  
}
```

```
atharva@quasar-2 Lab Assignment 5 % ./deletion
```

```
Enter size of array: 3
```

```
Enter the elements: 1 2 3
```

```
Original Array: 1 2 3
```

```
From which index do you want to delete? ->0
```

```
Update Array
```

```
1 2 %
```

Question4: Linear Search.

```
#include<stdio.h>
int main(){
    int n;
    int val;

    printf("\nLength of array: ");
    scanf("%d", &n);

    int arr[n];
    printf("\nEnter the elements of the array: ");
    for( int i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }

    printf("\nSearch item: ");
    scanf("%d", &val);

    for(int i = 0; i < n; i++){
        if (arr[i] == val){
            printf("%d was found at %d th index.", val, i);
        }
    }
    printf("\n");
}
```

```
atharva@quasar-2 Lab Assignment 5 % ./linearsearch
```

```
Length of array: 4
```

```
Enter the elements of the array: 1 2 3 4
```

```
Search item: 4
```

```
4 was found at 3 th index.
```

Question 5: Binary Search

```
#include<stdio.h>
int main(){
    int n, i, j;
    int temp;

    printf("\nEnter length of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("\nEnter the elements: ");
    for( i = 0; i < n; i ++){
        scanf("%d", &arr[i]);
    }

    printf("\nInitial Array\n");
    for( i = 0; i < n; i++){
        printf("%d, ", arr[i]);
    }

    for ( i = 0; i < n - 1; i++){
        for ( j = 0; j < n - 1 - i; j++){
            if (arr[j] > arr[j+1]){
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    printf("\nSorted Array\n");
    for( i = 0; i < n; i++){
        printf("%d, ", arr[i]);
    }
}
```

```
atharva@quasar-2 Lab Assignment 5 % ./bubble
```

```
Enter length of array: 6
```

```
Enter the elements: 23 04 33 6 9 88
```

```
Initial Array
```

```
23, 4, 33, 6, 9, 88,
```

```
Sorted Array
```

```
4, 6, 9, 23, 33, 88, %
```

Question 6: Sum and average of array elements.

```
#include<stdio.h>
int main(){
    int n, i;
    int sum;
    float avg;

    printf("\nEnter array length: ");
    scanf("%d", &n);

    int arr[n];

    printf("\nEnter array elements: ");
    for( i =0; i < n; i++){
        scanf("%d", &arr[i]);
    }

    printf("\nOriginal Array\n");
    for( i =0; i < n; i++){
        printf("%d, ", arr[i]);
    }

    for( i = 0; i < n; i++){
        sum += arr[i];
    }
    printf("\nSum = %d", sum-1);

    avg = sum / n;
    printf("\nAverage = %f \n", avg);
}
```

```
atharva@quasar-2 Lab Assignment 5 % ./sum_avg
```

```
Enter array length: 5
```

```
Enter array elements: 20 30 40 50 60
```

```
Original Array
```

```
20, 30, 40, 50, 60,
```

```
Sum = 200
```

```
Average = 40.000000
```

NOTE: Due to repetitive code for taking array length n as user input and then initializing an integer array of that length and then taking user input for all the elements of the array, I have created a header file to avoid the reptition.

array_utils.c	array_utils.h
<pre>1 #include <stdio.h> 2 #include "array_utils.h" 3 4 void inputArray(int arr[], int n) 5 { 6 for (int i = 0; i < n; i++) 7 { 8 printf("Enter element %d: ", i); 9 scanf("%d", &arr[i]); 10 } 11 } 12 13 void printArray(int arr[], int n) 14 { 15 for (int i = 0; i < n; i++) 16 { 17 printf("%d ", arr[i]); 18 } 19 printf("\n"); 20 } 21 }</pre>	<pre>1 #ifndef ARRAY_UTILS_H 2 #define ARRAY_UTILS_H 3 4 void inputArray(int arr[], int n); 5 void printArray(int arr[], int n); 6 7 #endif</pre>

Question7: Merge two arrays.

```
#include<stdio.h>
```

```
#include "array_utils.h"
```

```
int main(){
```

```
    int i, len1, len2;
```

```
    printf("\nEnter size of 1st array: ");
```

```
    scanf("%d", &len1);
```

```
    int arr_1[len1];
```

```
    printf("\nEnter size of 2nd array: ");
```

```
    scanf("%d", &len2);
```

```
    int arr_2[len2];
```

```
    printf("\nEnter elements in 1st array: ");
```

```
    inputArray(arr_1, len1);
```

```
    printf("\nEnter elements in 2nd array: ");
```

```
    inputArray(arr_2, len2);
```

```
    printf("\nArray 1: ");
```

```
    printArray(arr_1, len1);
```

```
    printf("\nArray 2: ");
```

```
    printArray(arr_2, len2);
```

```
    int arr_merged[len1 + len2];
```

```
    for(i = 0; i < len1; i++){
```

```
        arr_merged[i] = arr_1[i];
```

```
    }
```

```
    for(i = 0; i < len2; i++){
```

```
        arr_merged[len1 + i] = arr_2[i];
```

```
    }
```

```
    printf("\nMerged Array: ");
```

```
    printArray(arr_merged, len1 + len2);
```

```
}
```

```
atharva@quasar-2 Lab Assignment 5 % gcc merge.c array_utils.c -o merge
atharva@quasar-2 Lab Assignment 5 % ./merge
```

```
Enter size of 1st array: 3
```

```
Enter size of 2nd array: 4
```

```
Enter elements in 1st array: Enter element 0: 1
```

```
Enter element 1: 2
```

```
Enter element 2: 3
```

```
Enter elements in 2nd array: Enter element 0: 4
```

```
Enter element 1: 5
```

```
Enter element 2: 6
```

```
Enter element 3: 7
```

```
Array 1: 1 2 3
```

```
Array 2: 4 5 6 7
```

```
Merged Array: 1 2 3 4 5 6 7
```

```
atharva@quasar-2 Lab Assignment 5 %
```

Question 8: Find the largest and smallest element in an array.

```
#include<stdio.h>
#include "array_utils.h"
int main(){
    int n, i, j;

    printf("\nArray length: ");
    scanf("%d", &n);

    int arr[n];

    printf("enter the array elements: \n");
    inputArray(arr, n);

    printf("\nyour array: ");
    printArray(arr, n);

    //if we sort the array then the 0th element will by default
    // be the smallest and the n-1 th element will be largest.

    int temp;

    for(i = 0; i < n - 1; i++){
        for(j = 0; j < n - 1 - i; j++){
            if(arr[j] > arr[j + 1]){
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    printf("\nsorted array -> ");
    printArray(arr, n);

    int min = arr[0];
    int max = arr[n-1];

    printf("\n%d is the smallest and %d is the largest.\n", min, max);
}
```

```
atharva@quasar-2 Lab Assignment 5 % ./large_small
```

```
Array length: 4
```

```
enter the array elements:
```

```
Enter element 0: 23
```

```
Enter element 1: 64
```

```
Enter element 2: 33
```

```
Enter element 3: 12
```

```
your array: 23 64 33 12
```

```
sorted array -> 12 23 33 64
```

```
12 is the smallest and 64 is the largest.
```

```
atharva@quasar-2 Lab Assignment 5 %
```

Question 9: Take the union and intersection of two arrays.

```
#include<stdio.h>
#include "array_utils.h"
int main(){
    int i, j, n1, n2;
    int check;

    printf("\n1st Array lenght: ");
    scanf("%d", &n1);
    int arr_1[n1];

    printf("\nEnter elements: ");
    inputArray(arr_1, n1);

    printf("\n2nd Array lenght: ");
    scanf("%d", &n2);
    int arr_2[n2];

    printf("\nEnter elements: ");
    inputArray(arr_2, n2);

    printf("\ninitial arrays are: \n");
    printArray(arr_1, n1);
    printf("\n");
    printArray(arr_2, n2);

    int merge[n1 + n2];

    for(i = 0; i < n1; i++){
        merge[i] = arr_1[i];
    }
    for(i = 0; i < n2; i++){
        merge[n1 + i] = arr_2[i];
    }

    printf("\nUnion : ");
    printArray(merge, n1+n2);

    int inter[n1 > n2 ? n1 : n2];
    int k = 0;
```

```
for(i = 0; i < n1; i ++){
    check = 0;

    for(j = 0; j < n2; j++){
        if(arr_1[i] == arr_2[j]){
            check = 1;
            break;
        }
    }

    if(check){
        inter[k] = arr_1[i];
        k++;
    }
}

printf("\nintersection : ");
printArray(inter, k);
}
```

```
atharva@quasar-2 Lab Assignment 5 % ./mergeintersect
```

```
1st Array lenght: 4
```

```
Enter elements: Enter element 0: 9
```

```
Enter element 1: 8
```

```
Enter element 2: 7
```

```
Enter element 3: 6
```

```
2nd Array lenght: 3
```

```
Enter elements: Enter element 0: 4
```

```
Enter element 1: 5
```

```
Enter element 2: 6
```

```
initial arrays are:
```

```
9 8 7 6
```

```
4 5 6
```

```
Union : 9 8 7 6 4 5 6
```

```
intersection : 6
```

```
atharva@quasar-2 Lab Assignment 5 %
```

Question 10: Remove duplicates in an array.

```
#include<stdio.h>
int main(){
    int n, i, j, k;

    printf("\nArray length: ");
    scanf("%d", &n);

    int arr[n];

    printf("\nEnter the elements: ");
    for (i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }

    printf("\nThe original array: ");
    for (i = 0; i < n; i++){
        printf("%d ", arr[i]);
    }

    for( i = 0; i < n; i++){
        for ( j = i + 1; j < n; j++){
            if(arr[i] == arr[j]){
                for ( k = j; k < n - 1; k++){
                    arr[k] = arr[k + 1];
                }

                n--;
                j--;
            }
        }
    }

    printf("\nFinal array: ");
    for(i = 0; i < n; i++){
        printf("%d ", arr[i]);
    }
}
```

```
atharva@quasar-2 Lab Assignment 5 % ./duplicate
```

```
Array length: 3
```

```
Enter the elements: 1
```

```
1
```

```
2
```

```
The original array: 1 1 2
```

```
Final array: 1 2 %
```

```
atharva@quasar-2 Lab Assignment 5 %
```

Question 11: